Assignment No 7

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Title – Implement Backtracking search to solve the 8-Queens constraint satisfaction problem.

Program:-

```
N = 8

board = [-1] * N


def is_safe(row, col):
    for previous_row in range(row):
        previous_col = board[previous_row]


        # Same column
        if previous_col == col:
            return False


        # Same diagonal
        if abs(previous_row - row) == abs(previous_col - col):
            return False


    return True
```

```
def solve(row):
    # All queens are placed
    if row == N:
        return True

    # Try every column
    for col in range(N):
        if is_safe(row, col):
            board[row] = col

            # Place queen in next row
            if solve(row + 1):
                return True

            # Backtrack
            board[row] = -1

    return False

# Solve the problem
if solve(0):
    print("Solution found:")
```

```
    for row in range(N):
        for col in range(N):
            if board[row] == col:
                print("Q", end=" ")
            else:
                print(".", end=" ")
        print()
else:
    print("No solution found")
```

Output:-

```
Solution found:
Q . . . . . . .
. . . . Q . . .
. . . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . . Q .
. Q . . . . . .
. . . Q . . . .

Process finished with exit code 0
```